HANDS-ON 7 · DAY 2 / AFTERNOON · GROUP'S SHEET

Does any forecast beat the last value?

The case. The treasury team of Hands-on 6 buys an index fund once a month, on a day of its choosing, and now asks whether anything forecasts tomorrow's close of the Dow Jones better than today's close. Four simple forecasts can be built from the days before. One earns its place only if its errors are smaller than the last close's, on the same days.

DATA CARD · DOWJONES102425.CSV

What a row is. One trading day of the Dow Jones Industrial Average: 256 rows, no missing value. Used here: `Date`, read as a date on upload in Hands-on 6, and `Close`, the index at the day's end, in points. `Open`, `High`, `Low` and `Volume` are left aside.

The series forecast. `Close`, one trading day ahead, from the closes before it. The days before are trading days: a Monday's day before is the Friday.

How the rows were chosen. Every trading day from 1 October 2024 to 8 October 2025, with no weekend and no market holiday.

Its history. Downloaded for the previous edition of the module and used unchanged.

Source and licence. stooq.com, symbol `^DJI`, daily. No licence is recorded: used in class, never republished.

From panel 4. `bike-rentals.csv`, uploaded in Hands-on 6: Capital Bikeshare's daily rentals in Washington DC, every day of 2011 and 2012, so the same day last week is seven rows back; `Casual` counts the riders who were not registered. H. Fanaee-T (2013), UCI dataset 275, CC BY 4.0.

file	rows	period	series forecast	used in
`dowjones102425.csv`	256 trading days	10.2024 to 10.2025	`Close`, by `Date`	steps 1 to 3
`bike-rentals.csv`	731 days	2011 and 2012	`Casual`, by `Date`	panel 4

STEP 1 · PUT EACH DAY BESIDE THE FIVE BEFORE

1 ☐ **Lag the close**
On the Dow Jones dataset uploaded in Hands-on 6: **+ Recipe** → **Window**. In **Windows definitions**, order the rows by `Date`, ascending; in **Aggregations**, on `Close`, tick **Lag** with offsets `1,2,3,4,5`; **Run**.

Names of the lag-1 and lag-5 columns: _____

2 ☐ **Keep the days that have five days before**
On the lagged dataset: **+ Recipe** → **Prepare**, a step **Remove rows where cell is empty** on the lag-5 column, so that every forecast is scored on the same days.

Rows left: _____

STEP 2 · BUILD FOUR FORECASTS AND THEIR ERRORS

1 ☐ **Write each forecast's error**
In the same Prepare recipe, four steps **Formula**, one per forecast; ‹lag 1› stands for the lag-1 column's name, and so on. **Run**.

new column	the forecast of today's close	formula
`err_last`	the last close	`abs(Close - ‹lag 1›)`
`err_week`	the same weekday last week	`abs(Close - ‹lag 5›)`
`err_avg`	the average of the last five	`abs(Close - (‹lag 1› + ‹lag 2› + ‹lag 3› + ‹lag 4› + ‹lag 5›) / 5)`
`err_trend`	the last close plus the last change	`abs(Close - (2 * ‹lag 1› - ‹lag 2›))`

2 ☐ **Read each mean error**
On the prepared dataset, on each `err_` column: its header, **Analyze**, the **Mean**.

STEP 3 · SET EACH BESIDE THE LAST CLOSE

1 ☐ **Divide by the last close's error**
Below 1, a forecast beats the last close; this is the scaling MASE uses.

forecast	mean absolute error	÷ `err_last`'s	beats the last close?
the last close		1	(the benchmark)
the same weekday last week			
the average of the last five			
the last close plus the last change			

Does any forecast beat the last close? Why might the index behave so?

4 · BUILD THE SAME FORECASTS FOR THE CASUAL RIDERS

1 ☐ **Lag and score the riders**
On the bike rentals dataset uploaded in Hands-on 6: **+ Recipe** → **Window** ordered by `Date`, on `Casual`, **Lag** with offsets `1,2,3,4,5,6,7`; then **+ Recipe** → **Prepare**: **Remove rows where cell is empty** on the lag-7 column, and `err_last` = `abs(Casual - ‹lag 1›)`, `err_week` = `abs(Casual - ‹lag 7›)`, `err_avg` = the distance to the average of the seven lags; **Run**.

`Casual`	mean absolute error	÷ `err_last`'s	beats the last value?
the last value		1	(the benchmark)
the same day last week			
the average of the last seven			

5 · DECIDE

What do you tell the treasury team about tomorrow's close? Where did a forecast beat the last value, and what does that series have that the index lacks?